



BIOSOLIDS LAND APPLICATION SITE IDENTIFICATION FORM

*Authorized under 1994 PA 451, as amended. Completion of form is required.
Applicant may be subject to civil and/or criminal penalties for providing false information.*

Complete the following information and provide the attachments listed on the second page for each new or updated biosolids land application site. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e., new landowner).

For additional information, Biosolids Program Staff are listed at: https://www.michigan.gov/documents/deq/wrd-biosolids-staff_402800_7.pdf.

Facility Name City of Cadillac WWTP		NPDES Permit or COC Number MI0020257		Request: New Site / Site Update / Site transfer Site Updates	
Site ID Name (ex T/R/S – Owner (05N08E03-LB01) 21N09W27-WR01			Site location description (street name, nearest crossroad) 9720 E 48 RD		
County Wexford			Township Clam Lake		
(ex: NW, NE, SW, SE) NE 1/4		(ex: NW, NE, SW, SE) NW 1/4		Township (ex: 05N) 21N	Range (ex: 08E) 09W
Latitude (ex: N 42.09651) 44.19308 N		Longitude (ex: -81.09651) -85.53457 W		Method Used (i.e., GPS, topo) Google Earth	
Name of Owner William Runyon				Phone	
Street or P.O. Box 9720 E 48 RD			City or Town Cadillac		State MI
Zip 49601					
Is there a written agreement from the owner to land apply? ☒ Yes ☐ No (written agreement is required)					
Previous Landowner's Name (if site update) N/A				Property Purchase Date N/A	
Name of Lessee/Farmer (if different) John Rydquist				Phone 231-884-0719	
Street or P.O. Box 9897 S 43 RD			City or Town Cadillac		State MI
Zip 49601					
Site Type ☒ Agricultural ☐ Forest ☐ *Reclamation Site (separate approval is required) ☐ Lawn/Home Garden (requires EQ Biosolids)					
Soils Information Date of last soils analysis <u>9/27/21</u> *Phosphorous <u>64</u> units <u>ppm</u> Potassium <u>123</u> units <u>ppm</u> pH <u>6.7</u> Soil type(s) <u>Sandy</u> *300 lb/ac (or 150 ppm) Bray P1 maximum limit			Percent slope Highest percent slope of the site? ☐ 0-6% ☐ 6-12% ☒ *Higher (Requires an approved site reclamation plan)		
Expected Crops/Vegetation to be grown Crop/Vegetation: <u>Corn</u> Nitrogen Requirement: <u>135</u> lbs. N/acre Crop/Vegetation: _____ Nitrogen Requirement: _____ lbs. N/acre Crop/Vegetation: _____ Nitrogen Requirement: _____ lbs. N/acre					
Total acreage of site 75		Acreage used for crops 7.1		Acreage used for land application	
Does this site have a high potential for public exposure? ☐ Yes ☒ No			If agricultural, is the site tilled? ☐ Yes ☐ No ☒ NA		

CITY OF CADILLAC WASTEWATER TREATMENT PLANT

BIOSOLIDS RECYCLING PROGRAM PARTICIPATION

I hereby agree to participate in the City of Cadillac WWTP biosolids recycling program. I understand that biosolids will be applied to my land in accordance with federal EPA regulations and guidelines of the Michigan Department of Environmental Quality.

I understand that biosolids have been analyzed to be sure they can be applied to agricultural land in accordance with federal regulations and MDEQ guidelines. I understand that biosolids contain some heavy metals and that they must be below the EPA allowable limits for land application. I understand that the nutrients and/or the metal concentration may limit the frequency of biosolids application to my land.

My land is used for the production of agricultural crops. Information concerning crop and site restrictions has been explained to me. I agree to allow representatives of the MDEQ, Cadillac Waste Water Treatment Plant and the City's Injection contractor to have access to my land to obtain soil samples and for application of biosolids.

If I decide to cancel this agreement, I will give 60 days advance notice to the City of Cadillac WWTP.

Landowner: William Runyon Signed: William Runyon
Address: 9720 E 48th Date: 1-16-19
Township: Clam Lake Section: 27 Acres: 75
City WWTP Field Number: 21N9W27W01
If Rented, Farmer: John Rydquist
Address: 9897 S 43 Rd Cadillac MI 49601

Blue Rd. (M-55)
Lake City



www.kettunencenter.org
(231) 829-3421



CLAM LAKE T21N • R9W

07000

08000

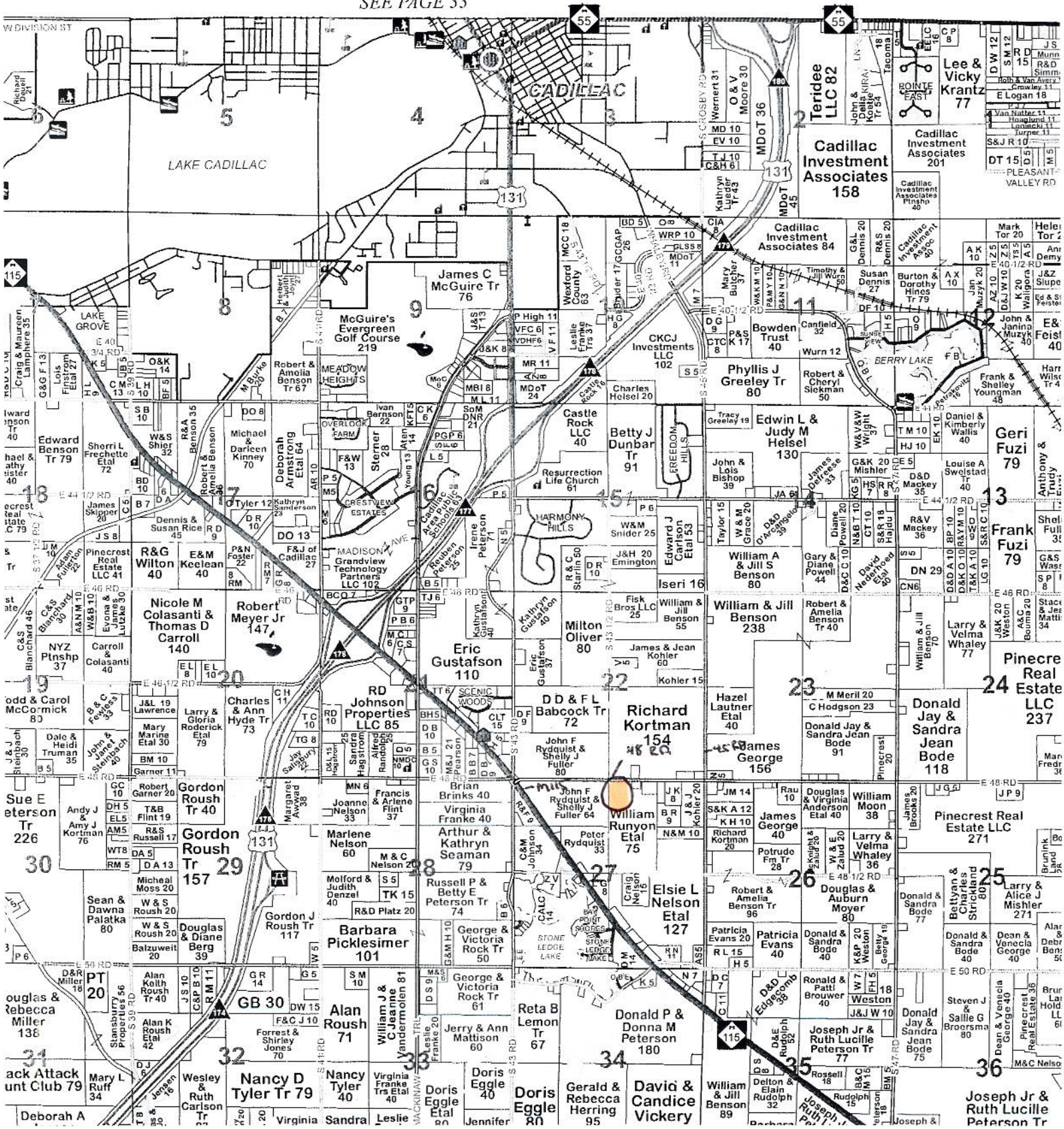
09000

10000

11000

1

SEE PAGE 33



**MICHIGAN STATE UNIVERSITY
SOIL AND PLANT NUTRIENT LABORATORY
EAST LANSING, MICHIGAN 48824-1325
(517) 355-0218**

SOIL TEST REPORT FOR:				CONSULTANT			
CITY OF CADILLAC 200 N. LAKE ST. CADILLAC MI 49601				WEXFORD COUNTY MSUE 401 N. LAKE STREET CADILLAC MI 49601 231-779-9480			
DATE	LAB #	COUNTY	Previous Crop	ACRES	FIELD ID	SOIL	TEXTURE
9/27/2021	260020	Wexford		07	21n09w27-wr01	Mineral	

SOIL NUTRIENT LEVELS			Below Optimum	Optimum	Above Optimum
¹ Soil pH	6.7	Lime Index 72.0			
² Phosphorus (P)	64	ppm	██████████	████████████████████	██
³ Potassium (K)	123	ppm	██████████	████████████████████	████
³ Magnesium (Mg)	99	ppm	██████████	████████████████████	███

ADDITIONAL RESULTS:					Optional Tests:						
Calcium (Ca) (ppm)	CEC (meq/100 g)	% of Exchangeable Bases			Micronutrients (ppm)					Organic Matter %	Nitrate-N ppm
		K	Mg	Ca	B	Cu	Mn	Zn	Fe		
847	5.4	5.9	15.3	78.8							

RECOMMENDATIONS:

Limestone: NONE

Tillage Depth: 9 inches

Target pH = 6.5

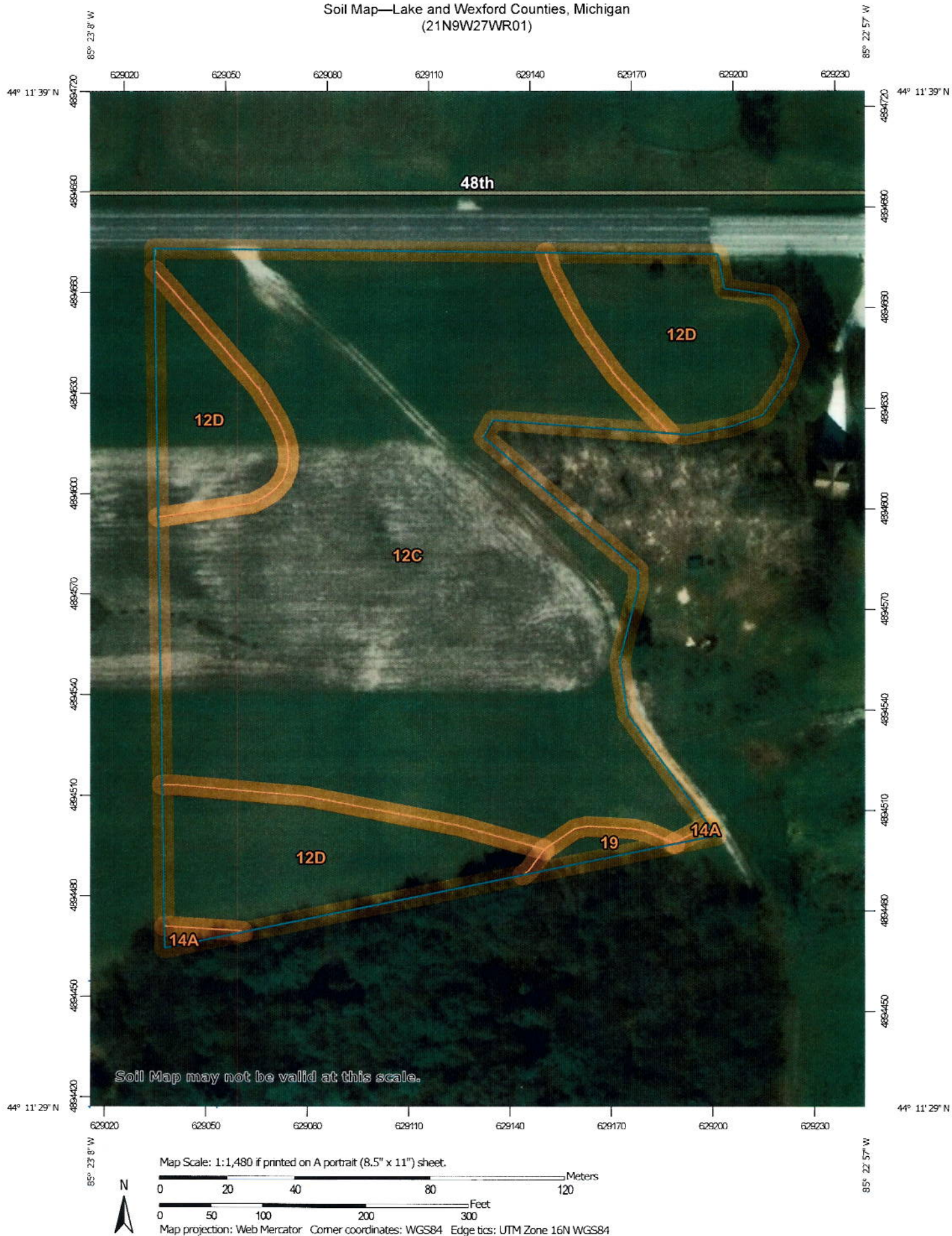
% Stand:

Plant Nutrients:					Micronutrient: (Optional)				
Year	Crop	Expected Yield	Nitrogen (lb N/A)	Phosphate (lb P ₂ O ₅ /A)	Potassium (lb K ₂ O/A)	Boron (lb B/A)	Manganese (lb Mn/A)	Zinc (lb Zn/A)	Copper (lb Cu/A)
1	Corn silage	18 ton	135	0	110	0.0			

Use a starter fertilizer high in nitrogen (30-40 lb N/A) when planting into heavy residue.

For questions about interpreting your soil results go to the following website: <http://msue.anr.msu.edu/experts> and contact the MSUE field crops agent in your area.

Soil Map—Lake and Wexford Counties, Michigan
(21N9W27WR01)



MAP LEGEND

Area of Interest (AOI)		Area of Interest (AOI)		Soils		Soil Map Unit Polygons		Soil Map Unit Lines		Other		Special Line Features		Special Point Features		Blowout		Borrow Pit		Clay Spot		Closed Depression		Gravel Pit		Gravelly Spot		Landfill		Lava Flow		Marsh or swamp		Mine or Quarry		Miscellaneous Water		Perennial Water		Rock Outcrop		Saline Spot		Sandy Spot		Severely Eroded Spot		Sinkhole		Slide or Slip		Sodic Spot	
Water Features		Streams and Canals		Transportation		Rails		Interstate Highways		US Routes		Major Roads		Local Roads		Background		Aerial Photography																																			

MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:15,800.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service
Web Soil Survey URL: websoilsurvey.sc.egov.usda.gov

Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Lake and Wexford Counties, Michigan
Survey Area Data: Version 11, Oct 6, 2017

Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: May 9, 2014—Sep 19, 2016

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Map Unit Legend

Map Unit Symbol	Map Unit Name	Acres in AOI	Percent of AOI
12C	Emmet-Montcalm complex, 6 to 12 percent slopes	5.1	71.5%
12D	Emmet-Montcalm complex, 12 to 18 percent slopes	1.9	27.1%
14A	Allendale loamy sand, 0 to 4 percent slopes	0.0	0.4%
19	Lupton muck, 0 to 1 percent slopes	0.1	1.0%
Totals for Area of Interest		7.1	100.0%